SUBHASHREE.M

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SUMMARY

Detail-oriented Embedded Systems fresher with hands-on experience in Embedded C, C programming, and microcontroller-based development on PIC18F4580. Skilled in UART, SPI, I2C, and CAN communication protocols, real-time peripheral interfacing (ADC, timers, interrupts, RTC, EEPROM), and Linux-based system programming. Built end-to-end embedded and software projects ranging from a CAN-based automotive dashboard to a real-time vehicle event-logging system. Seeking an entry-level Embedded Software Engineer / Firmware Engineer role to apply strong C and embedded fundamentals to real-world product development.

TECHNICAL SKILLS

- Programming Languages: C, Embedded C
- Embedded Systems: ADC, Timers, Interrupts, EEPROM, RTC Interfacing, CLCD Interfacing, Peripheral Interfacing, Real-Time Data Acquisition
- Microcontrollers: PIC18F4580, PIC Microcontroller Family
- Communication Protocols: UART, SPI, I2C, CAN
- Operating Systems: Linux (Shell Scripting, Process Management, System Calls)
- Tools & Concepts: Git, GitHub, Data Structures & Algorithms, File Handling, Modular Programming
- Currently Learning: ARM Cortex-M Architecture, STM32

WORK EXPERIENCES

Embedded system Trainee - Emertxe Information technology

•Built a strong foundation in C and Embedded C through structured, hands-on coding practice and lab exercises.
•Programmed microcontroller peripherals including ADC, timers, and interrupts to support real-time embedded applications.
•Implemented and tested UART, SPI, I2C, and CAN communication protocols for reliable embedded data exchange.
•Applied data structures and Linux-based development workflows to embedded and systems programming exercises.

EDUCATION

Government college of Engineering-Bodi - **B.E** Electrical and Electronics Engineering - 2021-2025|**CGPA**:7.76

HSC — Vellalar Matric Higher Secondary School, Erode| 88.8%

SSLC — Vellalar Matric Higher Secondary School, Erode| 92.5**%**

CERTIFICATIONS & PROFESSIONAL DEVELOPMENT

·Presented a technical paper at Government College of Engineering (GCE), Bodi, demonstrating strong technical communication skills.

·Presented a project at the Dr. Abdul Kalam Awards – Inter-College Project Presentation, showcasing practical implementation and problem-solving skills in a competitive environment.

·Completed Electric Vehicle (EV) Boot Camp at Sri Ramakrishna Institute of Technology (SRIT), Coimbatore, with certification.

.Received recognition in a MathWorks MATLAB-based technical activity for designing and presenting a structured diagram, demonstrating analytical thinking and visualization skills.

PROJECTS

CAR BLACKBOX

Technologies: Embedded C, UART, I2C, RTC, EEPROM

- Developed an automotive black box system to monitor and record critical vehicle parameters and events.
- Implemented real-time data logging using RTC and EEPROM for storing event information.
- Integrated peripherals such as UART, I2C and display interfaces for data communication and monitoring.

CAN AUTOMOTIVE DASHBOARD

Technologies: Embedded C, CAN

- Developed an automotive dashboard to monitor and display vehicle parameters received through the CAN bus.
- Implemented CAN communication for transmitting and receiving real-time vehicle data between nodes.
- Interfaced the dashboard with display and input peripherals for effective parameter monitoring.

MINISHELL

Technologies: C, Linux, System Calls

- Developed a Linux-based command-line shell using C to execute and manage user commands.
- Implemented process creation and execution using system calls such as fork(), exec() and wait().
- Handled built-in commands, background processes and command-line input processing.

LEXICAL ANALYZER

- Developed a lexical analyzer in C to identify and classify tokens such as keywords, operators, identifiers and constants.
- Implemented source-code parsing by handling whitespace, comments and different token types.
- Developed a Source-to-HTML conversion module to generate an HTML representation of C source code.

INVERTED SEARCH

Technologies: C, File Handling, Linked Lists, Hashing

- Developed an inverted search system to index and search words across multiple text files.
- Implemented hashing and linked-list-based data structures to efficiently store and retrieve word occurrences.
- Implemented file-based searching and indexing operations with support for updating and displaying search results.

APC

Technologies: C, Data Structures, Linked Lists

- Developed a calculator in C to perform arithmetic operations on integers exceeding the standard data-type limits.
- Implemented addition, subtraction, multiplication and division using linked-list-based representation of large numbers.
- Handled sign, carry/borrow and large-number operations to produce accurate results without integer overflow.

MP3-TAG READER

Technologies: C, File Handling, Structures, Pointers

- Developed an MP3 Tag Reader in C to read and display metadata such as title, artist, album, year and genre.
- Parsed ID3 tag information from MP3 files using file-handling operations and binary data processing.
- Implemented command-line options to view and modify MP3 tag information efficiently.